09D — Deep-Dive Feature Audit: Combat Engine & Tactical Simulation (Part 4 of 6)

Multi-Entity Battlefield Units: Summons (`W.foeById` Leak & Stepping Bug), `1-HP` Shadow Clones vs AoEs, Bonded Pet Assists, Weapon Avatars, Founder Phantoms & On-Map Party Synergies

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` in Google Drive `Js2 new`)
- **Target Audience**: Autonomous AI Developers (**GPT** & **Gemini**)
- **Core Game Philosophy**: **100% Emergent, Unified Multi-Entity Combat**. Summoned avatars, ninja shadow clones, spirit beasts, guardian constructs, ancestral phantoms, and cross-path party formations must work seamlessly in **real-time on-map battles**—not just in off-map text formulas or player-only UI buttons!
- **Master Roadmap for Category 09 (`09A–09F`)**:
  - `09A`: Core Damage Formula (`dmg = hpMax(d) * frac`), 12-Stat Split-Brain, Elemental Matchups & Behavioral Mastery Traits
  - `09B`: Spatial Battlefield Geometry, `NPC` vs `foe` Movement Asymmetry, `24`-Tile `ROW_BODYBLOCK` Bug, Interception & AoE Selectors
  - `09C`: Action Economy, Charged/Channeling Multiplier Loss, `statusTick` DoT Loops, Shield Overwrite, Acupoints & Boss Anatomy Breaker
  - **`09D` (This Document)**: **Multi-Entity Battlefield Units — Summons, Shadow Clones, Constructs, Pet Assists, Gate Beasts, Weapon Avatars, Founder Phantoms & Party Synergies (`combat.js`, `sim.js`, `sect.js`)**
  - `09E`: Fight Modes, Stakes, Morale/Yield/Retreat, 7-Layer Cheat-Death Stacking, Raid Core-Breach AI, Self-Defense Jail Bug & Heavenly Tribulations
  - `09F`: Grand Finale — Holistic Overview of the Entire Combat Ecosystem

---

1. `[CMB-SUM-01]` Four Concrete Bugs in Summoned Units (`arch === 'summon'`, `js/combat.js:765, 899–924` & `js/sim.js:342`)!

1. **Permanent `W.foeById` Memory Leak Across Multi-Decade Saves (`combat.js:765` vs `920`)**:

- Line `920` registers every summoned unit into `W.foeById[sId] = summonUnit` (`sId = 'SUM_' + a.id + '_' + W.t`).
- Line `765` (`C.conclude`) filters `W.foes = W.foes.filter(x => !x.isSummon || !x.dead)`—**but NEVER deletes `W.foeById[x.id]`**! Over thousands of battles, dead summon objects accumulate infinitely in `W.foeById`!
- **Fix**: Before filtering dead summons/avatars/phantoms out of `W.foes` in `C.conclude` (and on `dur <= 0` expiry in `C.foeStep`), always `delete W.foeById[x.id]`!

2. **Summons Depend on `EXPED.raidTick(W)` Instead of `COMBAT.tick(W)` (`sim.js:342–344`)**:
   - In standalone `COMBAT.tick(W)` loops or non-raid contexts where `EXPED.raidTick` skips non-raid stepping, `C.foeStep` is never invoked for friendly summons!
   - **Fix**: Inside `COMBAT.tick(W)`, step any living `isSummon` / `isWeaponAvatar` / `isFounderPhantom` unit in `f.A` or `f.B` whose `u._lastStepTick !== W.t` (marking `u._lastStepTick = W.t` so `raidTick` never double-steps them!).
3. **Every Summon Uses a Generic Tier-1 `{ M: 0.85, cd: 8 }` Stick Attack & Bleeds Like Flesh (`line 918`)**:
   - Inherit `tier: p.tier || 1`, `element: p.element`, and scale `M = 0.80 + 0.08 * (p.tier || 1)` on the summon's attack, and grant inorganic summons (`construct`, `sword_phantom`, `skeleton`, `golden_soldier`) immunity to biological `bleed` and `poison`!
4. **Unimplemented `destroySummons: true` on `4E-04` Nine Suns Apocalypse (`combat.js:1258` vs `1365–1435`)**:
   - `NINE_SUNS_APOCALYPSE` declares `destroySummons: true`, **yet `triggerElementalReaction` never checks `spec.destroySummons`**! Add `if (spec.destroySummons && f) { C.enemies(W, f, a.id).filter(u => u.isSummon).forEach(s => C.applyDamage(W, f, a, s, s.hp + 999, spec.nameTh)); }`!

---

2. `[CMB-CLONE-02] 1-HP` Ninja Shadow Clones (`js/combat.js:356–366`) Absorb `100%` of Screen-Wide AoE Cataclysms Unscathed!

- **Exact Bug (`combat.js:356–366`)**:
  - Whenever `d._shadowClones > 0`, `C.applyDamage` decrements `1` clone and immediately `return 0;`—**even when hit by an AoE meteor storm, dragon breath, or `isAoeMultiTarget` ultimate**!
  - This directly contradicts the `NINPO_SHADOW` design rule (*"ร่างแยกเงา 1 HP แพ้ทางสกิลโจมตีวงกว้าง AoE อย่างหนัก"*).

- **Exact Fix**: When the incoming hit carries `isAoeMultiTarget` (or `aoeSplash` elemental reaction):
    - Shatter **all** active `1-HP _shadowClones` at once (`d._shadowClones = 0; clearBuff('shadowClone')`), reducing the AoE damage reaching the real body by `30%` rather than `100%`, and apply a brief `2`-tick `'exposed'` opening when all clones shatter simultaneously!

---

3. `[CMB-PET-03]` Bonded Spirit-Beast Pack Assists (`js/combat.js:583–615`) Scale Off a Scholar Owner's Weak Physical `sa.atk`!

- **Exact Bug (`combat.js:601`)**:

```
JavaScript
const pRes = C.strike(W, f, a, d, { M: assistM, type: 'p', ..., isPetAssist: true
});
```

    - Because `C.strike` evaluates `a` (`the human owner`) with hardcoded `type: 'p'`, a `HEART_SOUL` / `XIAN` / `SAGE_SUPPORT` / `ASTRO_DIVINATION` mage (`matk = 1,400`, `atk = 260`) whose Tier-6 **Mythic Astral Kirin** lunges at an enemy deals damage scaled off **`260` physical `atk`**!
- **Exact Fix**:
    - Pass `type: sa.matk > sa.atk * 1.1 ? 'm' : 'p'` (or pass an explicit `atkOverride` blending `Math.max(sa.atk, sa.matk) * 0.70 + beastBaseAtk(pt.tier, pt.level) * 0.30` and `element: pt.element`) so Spirit Beasts hit with power worthy of their own Tier and Element regardless of whether their master is a scholar or a brawler!

---

4. `[CMB-SYN-04]` Party Formation Synergies (`evaluateFormationSynergy` & `CROSS_PATH_SYNERGIES`, `js/combat.js:1265–1340`) Have `0.00%` Effect in Real On-Map Combat + Missing `16` Paths & Dormant `invokeAncestralPhantom`!

1. **On-Map vs Off-Map Synergy Blackout (`combat.js:1265–1340`)**:

- Full-file inspection of `combat.js` confirms `C.evaluateFormationSynergy` (`8` classic party auras) and `C.evaluateCrossPathSynergies` (`16` Grand Dao pairings) **are NEVER called inside `C.strike`, `C.applyDamage`, or `C.tick`**!
- They apply **only** in off-map text resolvers (`Sanctums`, `Pagoda`, `Sieges`). When the exact same 5-person party fights live on the map in a **Sect Raid (`raid`)** or **Boss Hunt (`wild`)**, all `24` party synergies grant **`0.00%`** power and **`0.00%`** mitigation!
- **High-Performance O(1) Fix (`Zero Tick Overhead!`)**: Cache `f._formationSynA` and `f._formationSynB` on `f` once every `4` ticks (`when f.A.length >= 2 || f.B.length >= 2`), applying a gentle bounded on-map aura (`1 + (syn.powerMultiplier - 1) * 0.5` offense, `1 - syn.damageMitigation * 0.5` mitigation) inside `C.strike`—and expand `CROSS_PATH_SYNERGIES` to include the `16` `v2.1–v2.3` Grand Dao Paths (`e.g. TIDE_VOYAGER + CELESTIAL_WEATHER`, `ARENA_GLORY + BODY`, `SPIRIT_HUSBANDRY + VENOM_GU`, `LORE_ARCHIVIST + ASTRO_DIVINATION`, `ANCESTRAL_RITUAL + DIVINE_SUMMONER`, `FORTUNE_GAMBLER + MERCANTILE_DAO`, `DEVOURING_FLESH + DEMON`, `VOID_HERETIC_PACT + HEART_SOUL`)!

2. **Dormant `ANCESTRAL_PAVILION` & `invokeAncestralPhantom` (`v340` Phase 7)**:
   - Currently `enshrineAncestralRelic` and `invokeAncestralPhantom` (`combat.js:1300`) are **100% UI-button locked**!
   - **Autonomous Crisis Trigger**: Let `scripture`/`treasury` enshrine duplicate retired T5+ relics onto the Ancestral Altar, and during a **critical Sect Raid (`f.kind === 'raid'`)** when `breachTicks >= 12` or `>= 40%` of defenders have fallen and `incensePoints >= 50`, let the **Sect Master / Ancestral Elder autonomously invoke the Founder Phantom** to rally the sect!

---

5. Mandatory Verification Suite Additions (`09D`)

1. **Summon `W.foeById` Cleanup & Standalone Stepping Test**: Cast `summon` in a standalone `COMBAT.tick` duel (`no raidTick`), assert the summon attacks at least once, and assert `W.foeById[summonId] === undefined` after `C.conclude`!
2. **Shadow Clone AoE Counterplay Test**: Give a ninja `3` `_shadowClones`, hit them with an `isAoeMultiTarget: true` skill, and assert all `3` clones shatter and reduced AoE damage reaches HP.
3. **Live On-Map Cross-Path Synergy & Crisis Founder-Phantom Test**: Put `VOID_SPACE` + `CHRONO_EPOCH` allies in a live on-map fight and assert `SYN_SPACETIME_CONTINUUM` boosts real `C.strike` output; simulate a near-breach `raid` with `incensePoints >= 100` and verify autonomous `invokeAncestralPhantom`.